Evolving Boxes for Fast Vehicle Detection

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[3]University of Washington

Outline

- Introduction and Motivation
- Our Framework
  - System Framework
  - Network Training
- Experiments
- Conclusion

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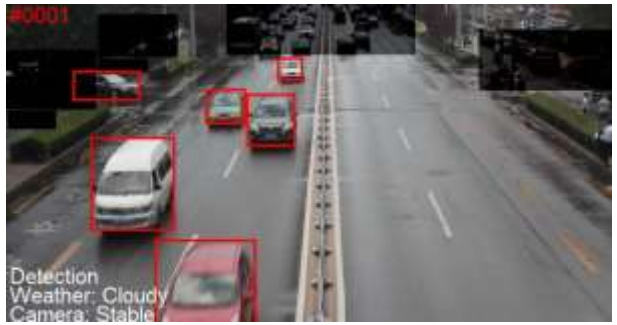
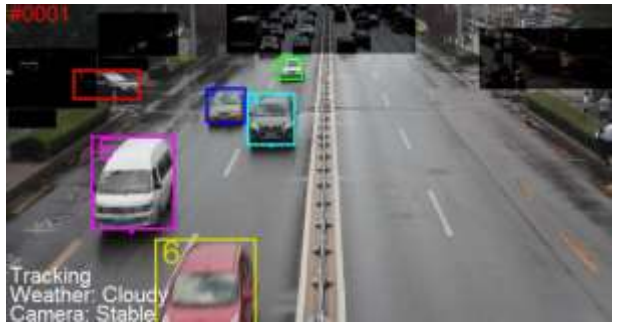
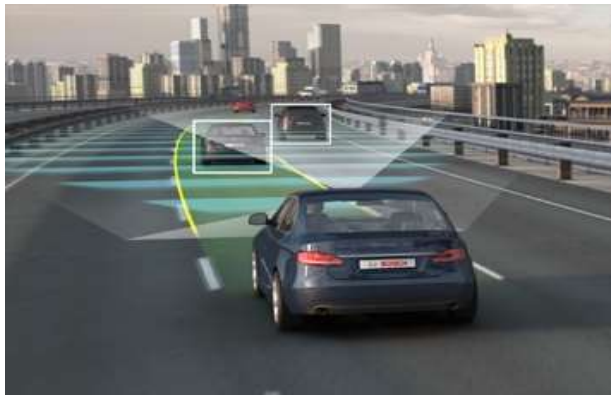
Background

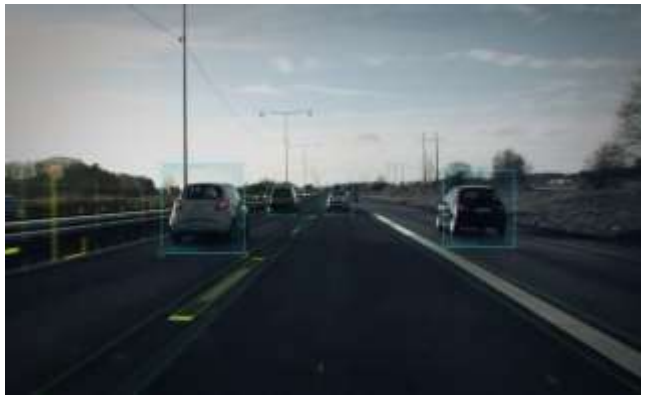
- Vehicle Detection
  - Traffic Surveillance System





  - Autonomous Driving System





Challenges

Heavy occlusions

Multiple scales

L. Wen, D. Du, Z. Cai, Z. Lei, M.-C. Chang, H. Qi, J. Lim, M.-H. Yang, and S. Lyu, “DETRAC: A new benchmark and protocol for multi-object detection and tracking,” *arXiv preprint arXiv:1511.04136*, 2015.

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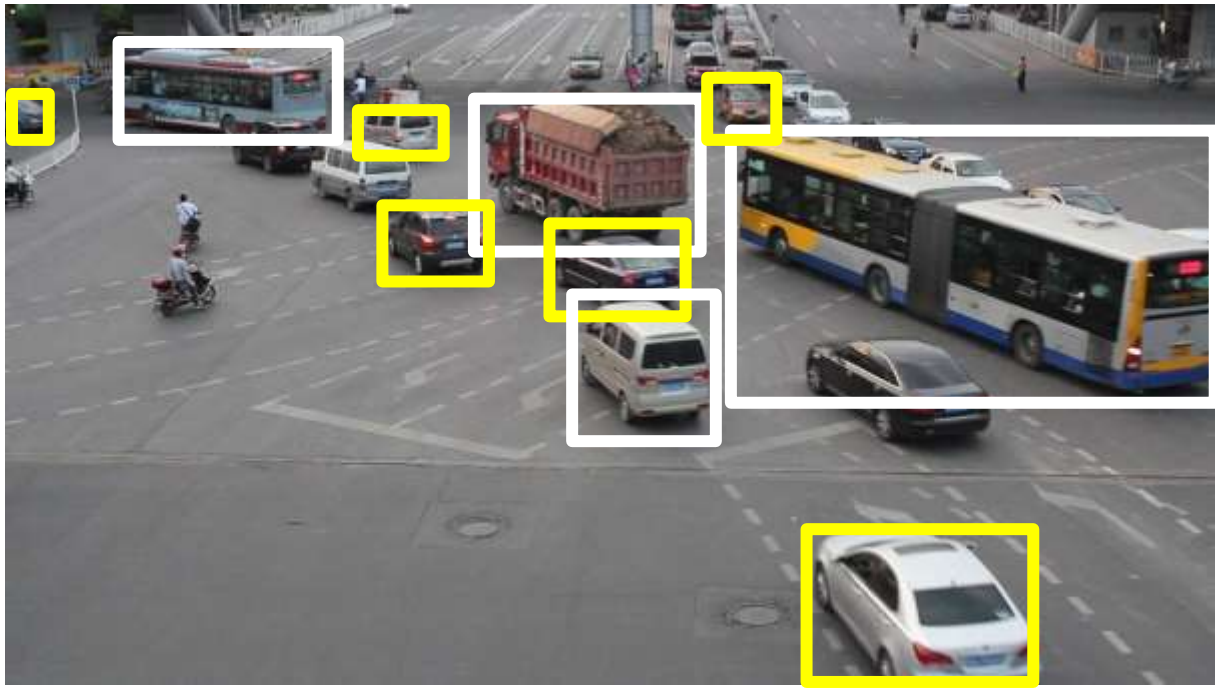

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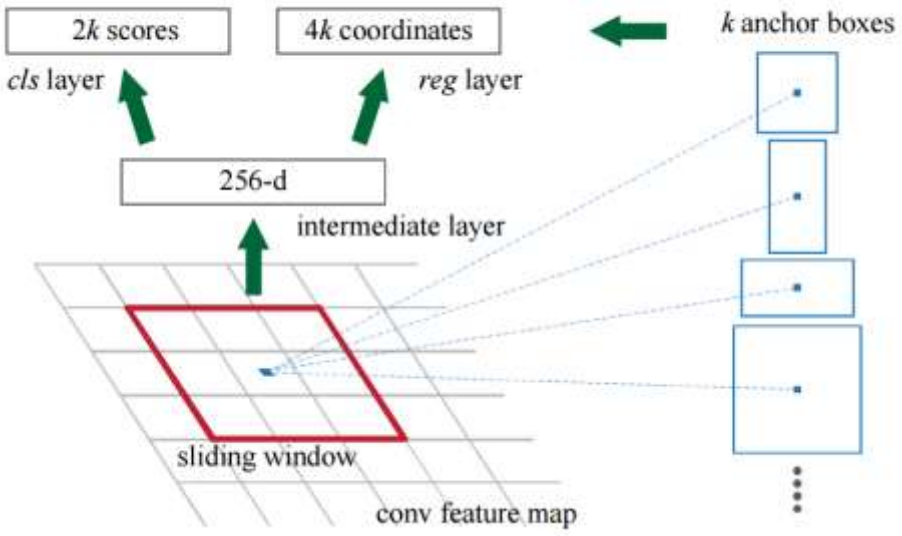
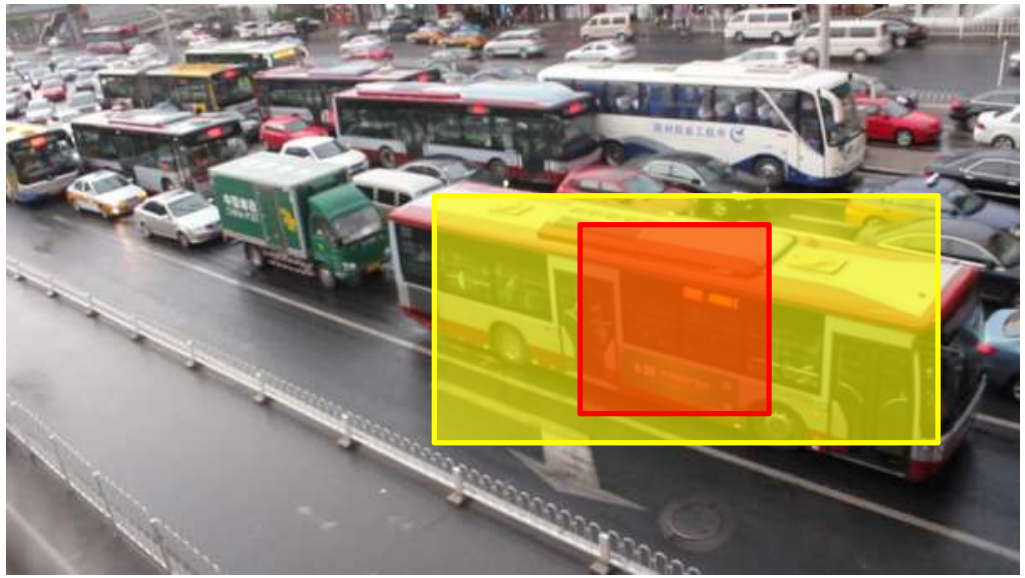
Motivation

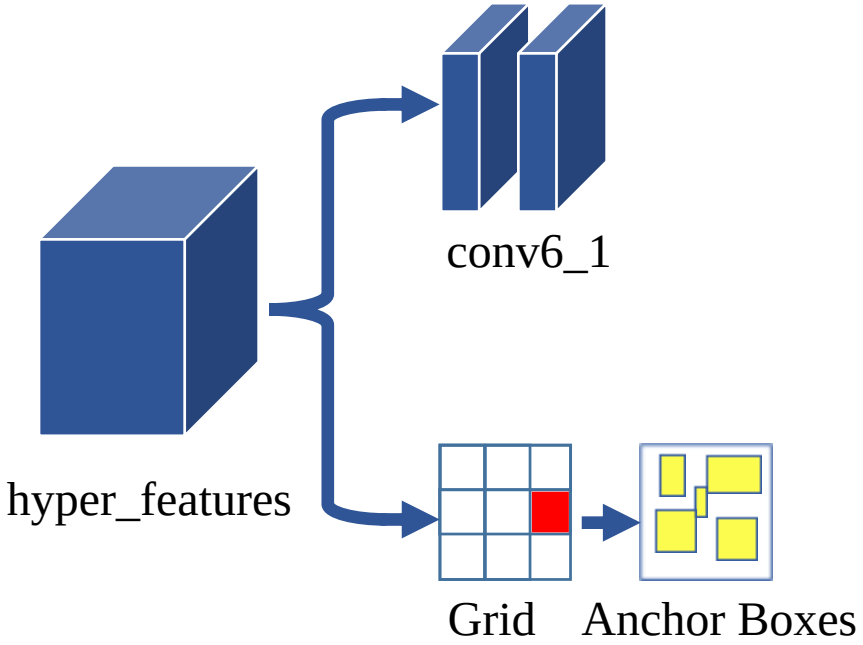


- Anchor Boxes are predicted and regressed based on global features:
  - Discrimination between different proposals is weak
  - Decrease the accuracies of scores and bounding boxes

S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *NIPS*, 2015.

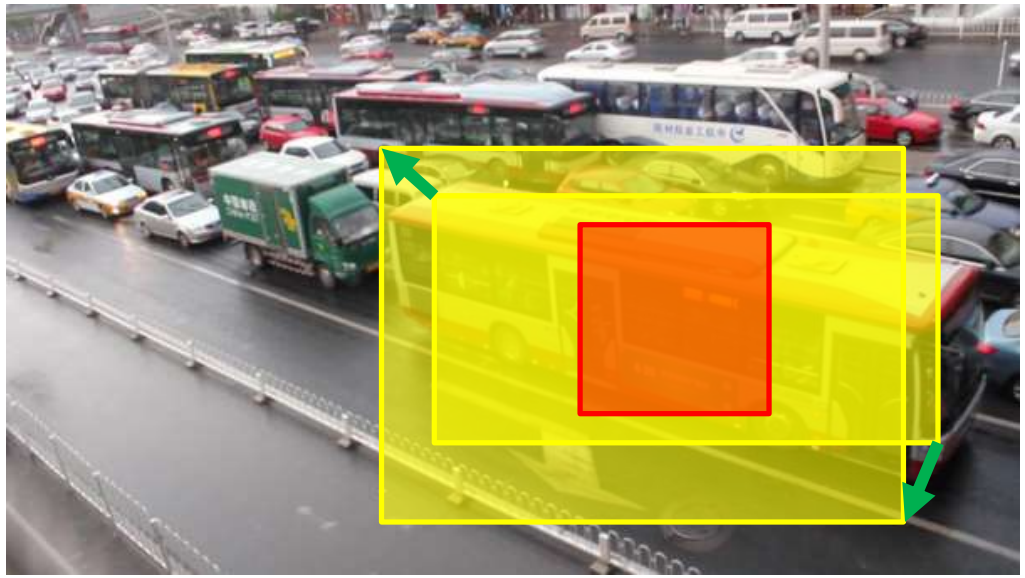
Motivation

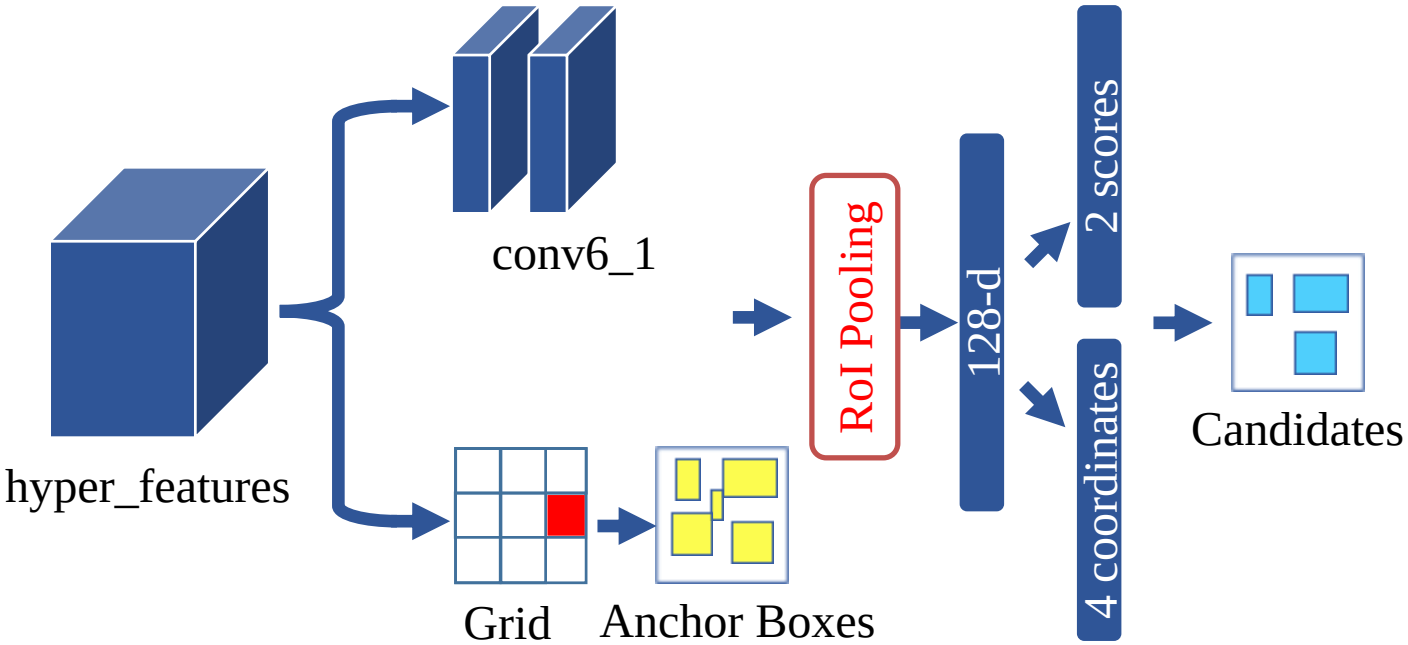




- Evolving the anchor boxes to predict and regress on local features:
  - Strengthen the discrimination of features between proposals
  - Facilitate the scores and position regression

Motivation

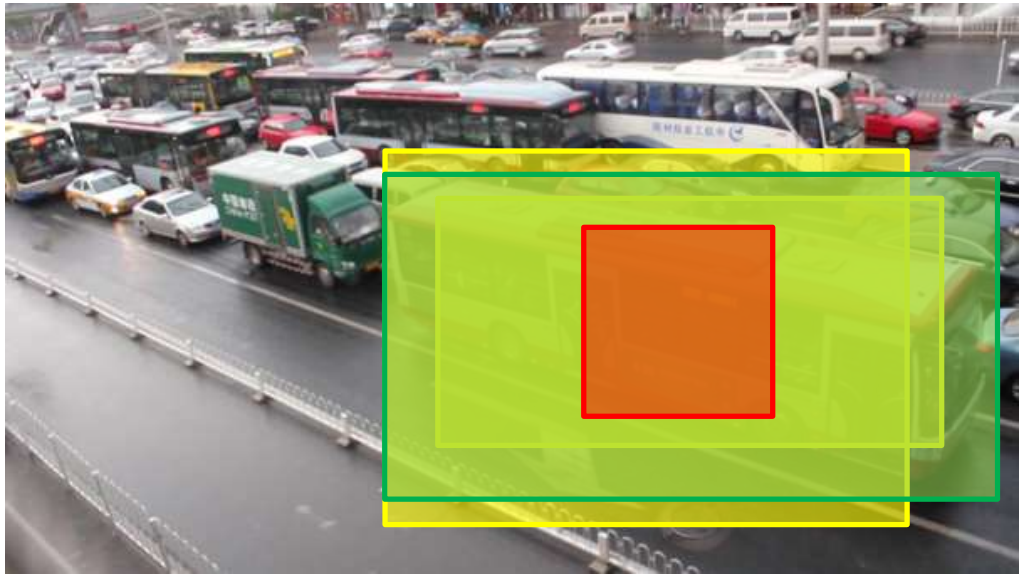


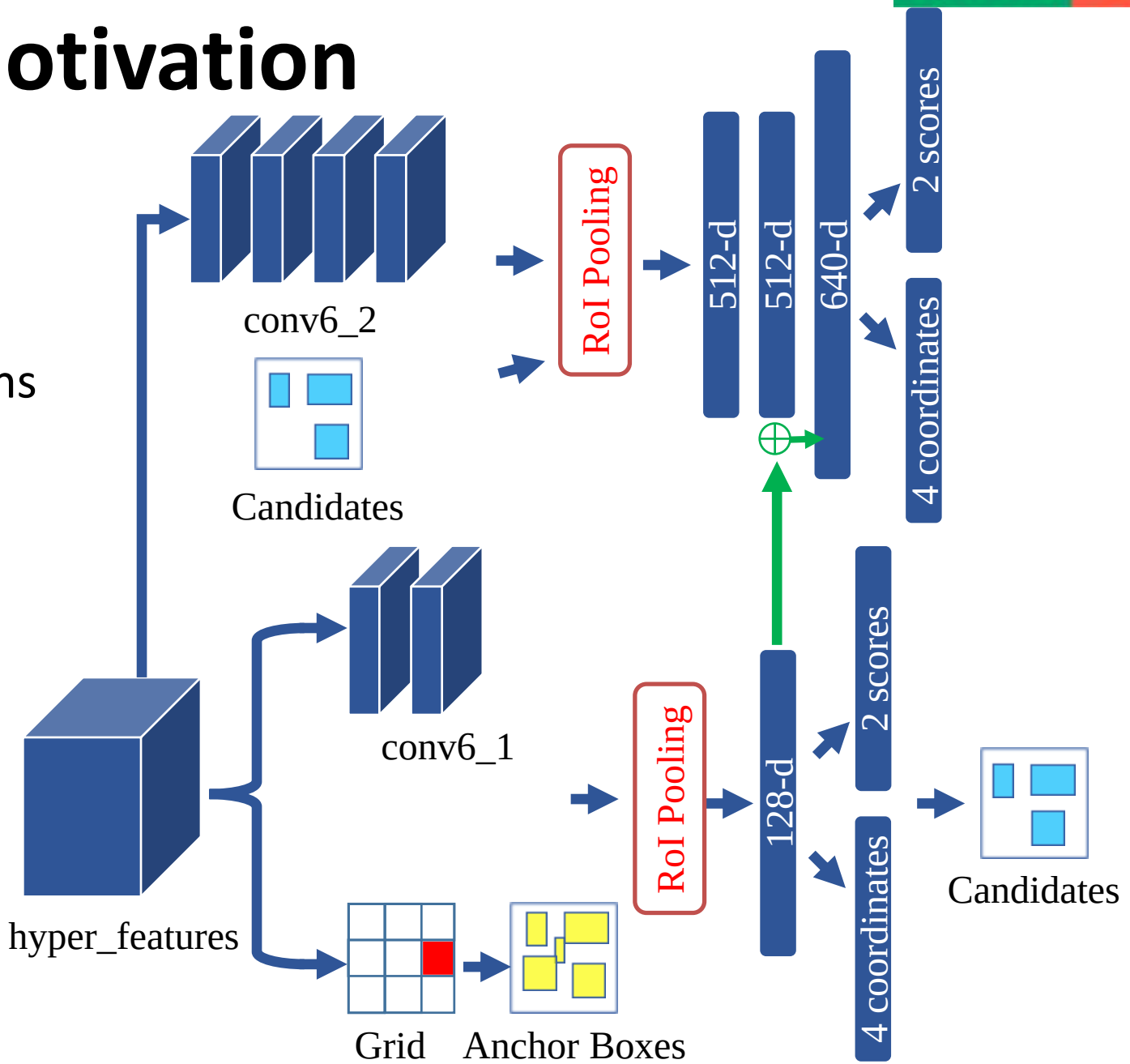


- Evolving the anchor boxes to predict and regress on local features:
  - Strengthen the discrimination of features between proposals
  - Facilitate the position regression

Motivation

- Feature Concatenation for evolving boxes:
  - enrich the feature representations





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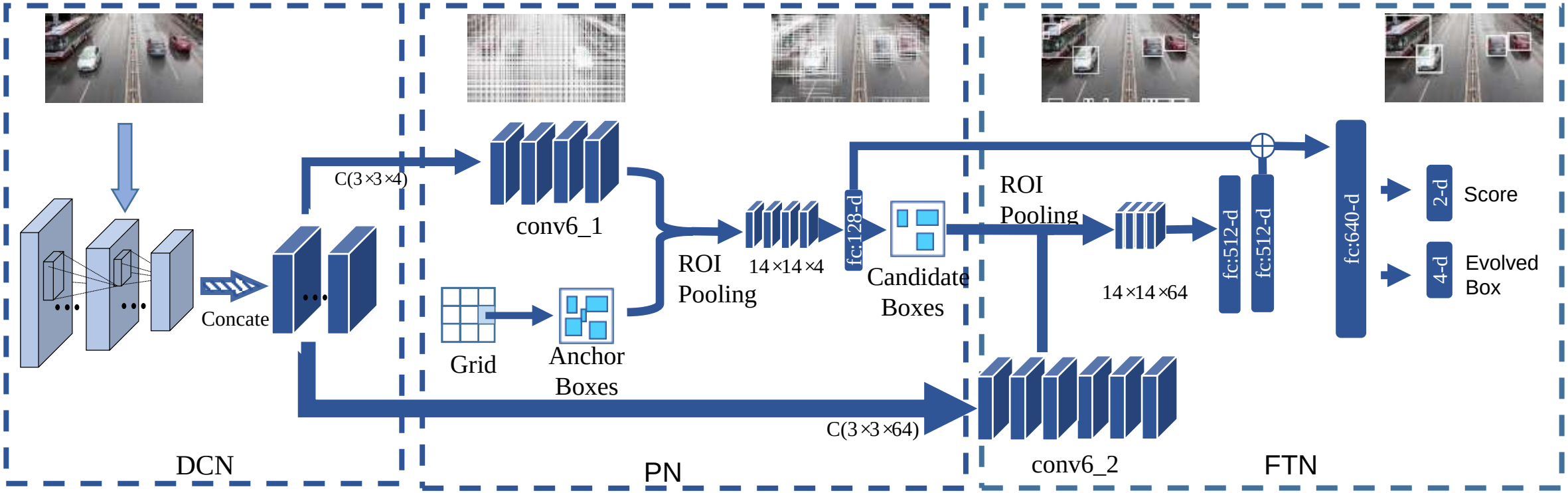
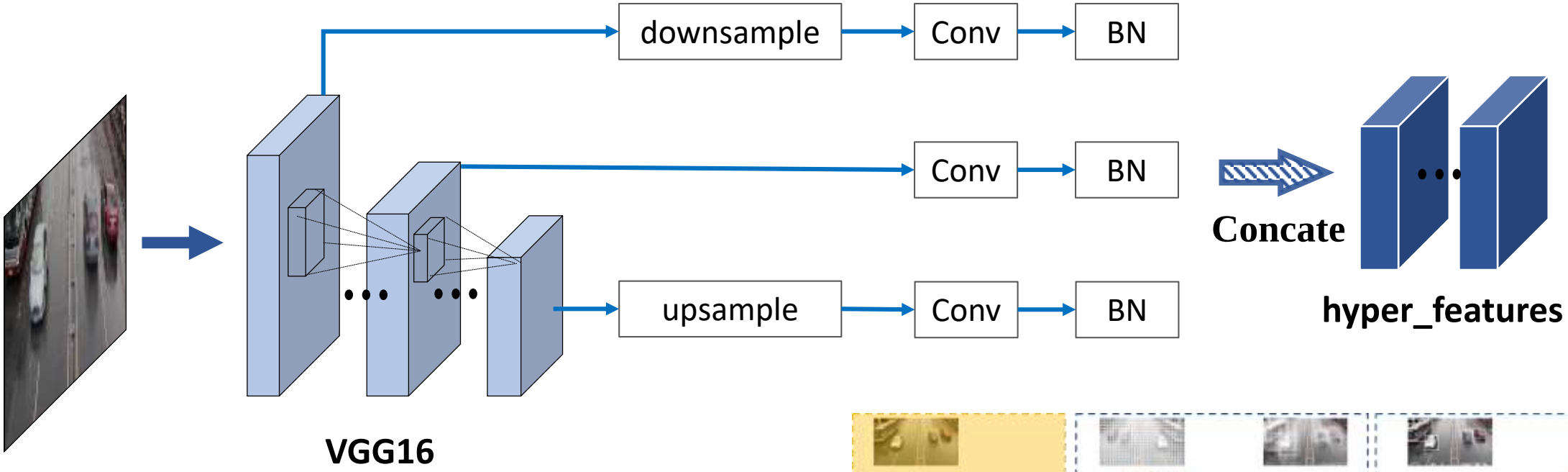
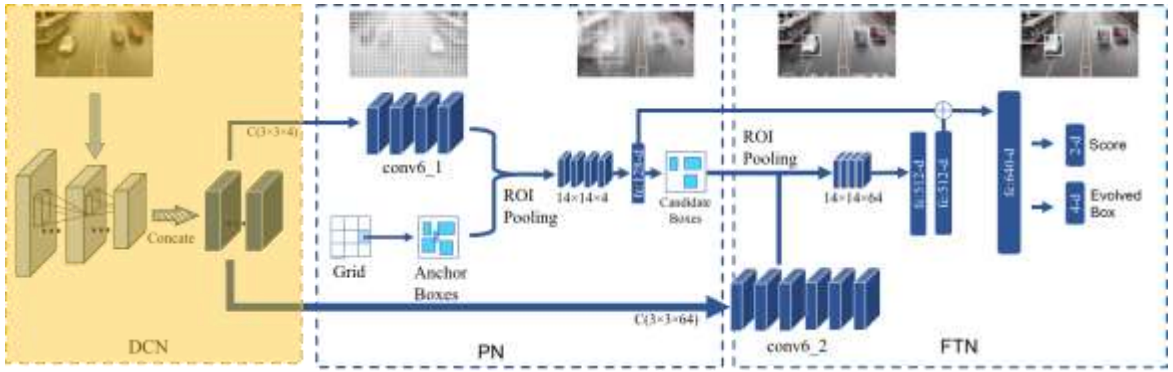
Framework

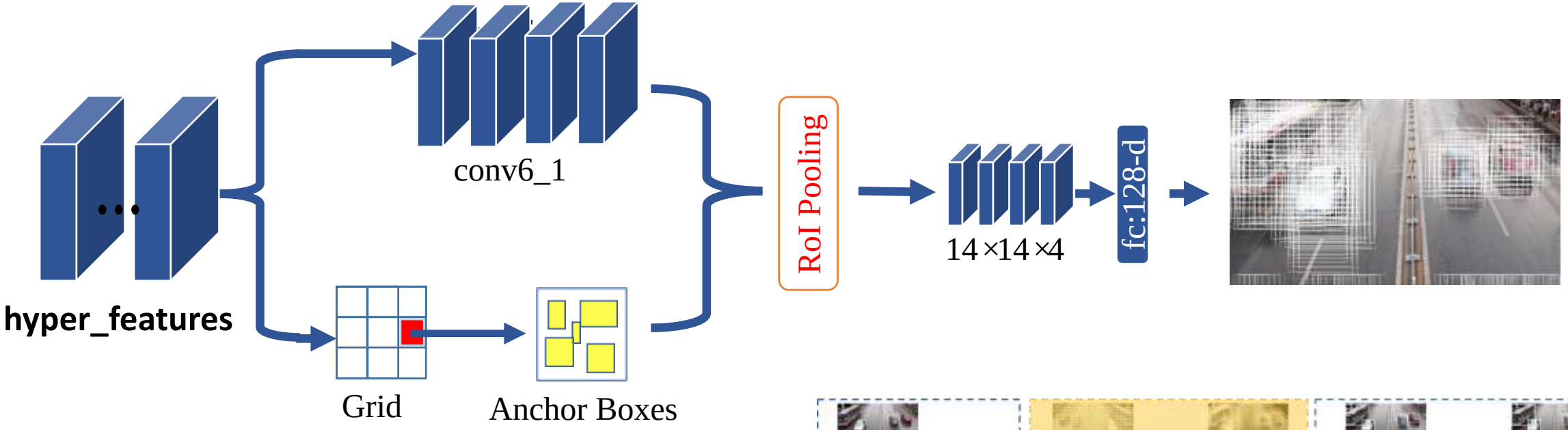

C(3×3×4)
conv6_1
Concate
Grid
Anchor Boxes
ROI Pooling
14×14×4
fc:128-d
Candidate Boxes
ROI Pooling
14×14×64
fc:512-d
fc:512-d
fc:640-d
2-d
Score
4-d
Evolved Box
C(3×3×64)
conv6_2
DCN
PN
FTN

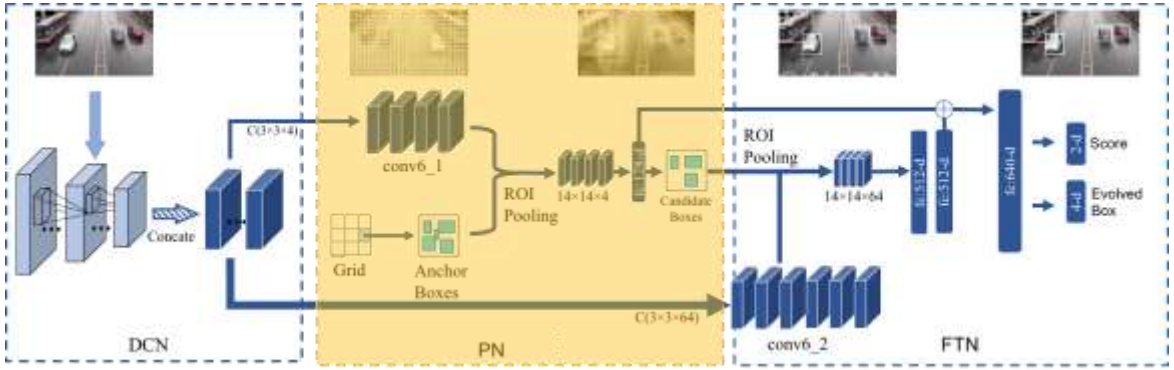
Deep Convolutional Network



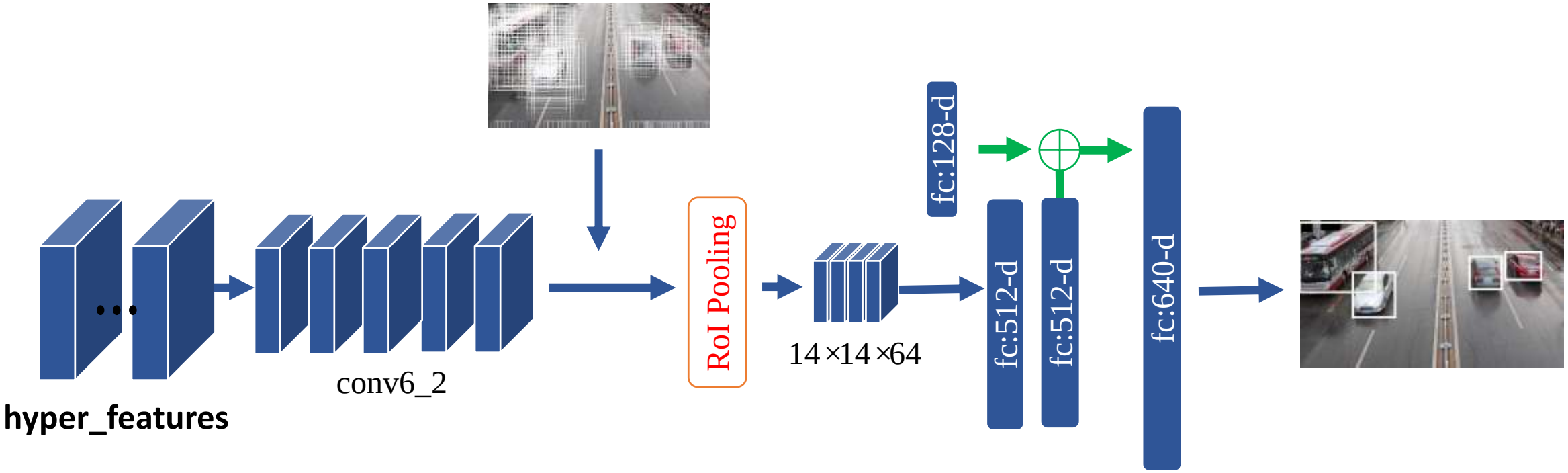


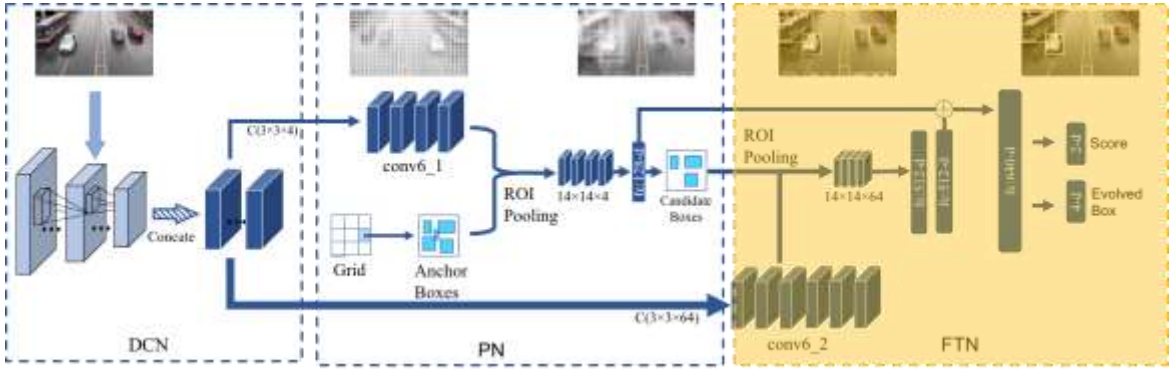
Proposal Network





Fine-tuning Network





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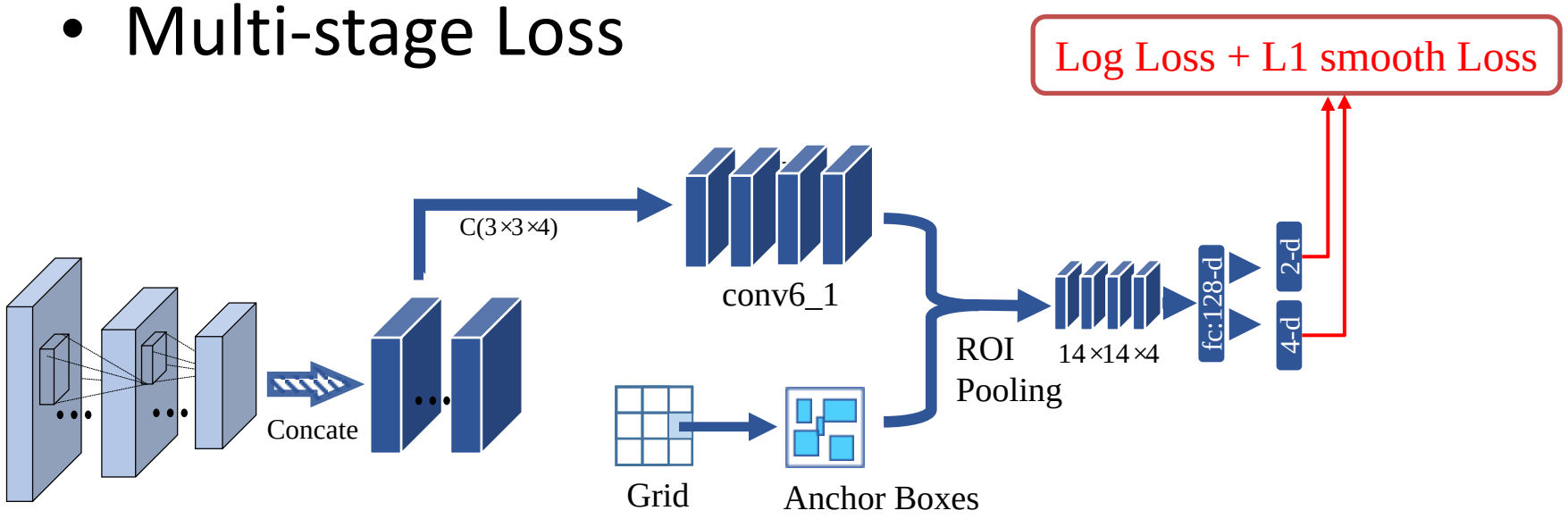
Network Training

- End-to-end training
- Multi-stage Loss



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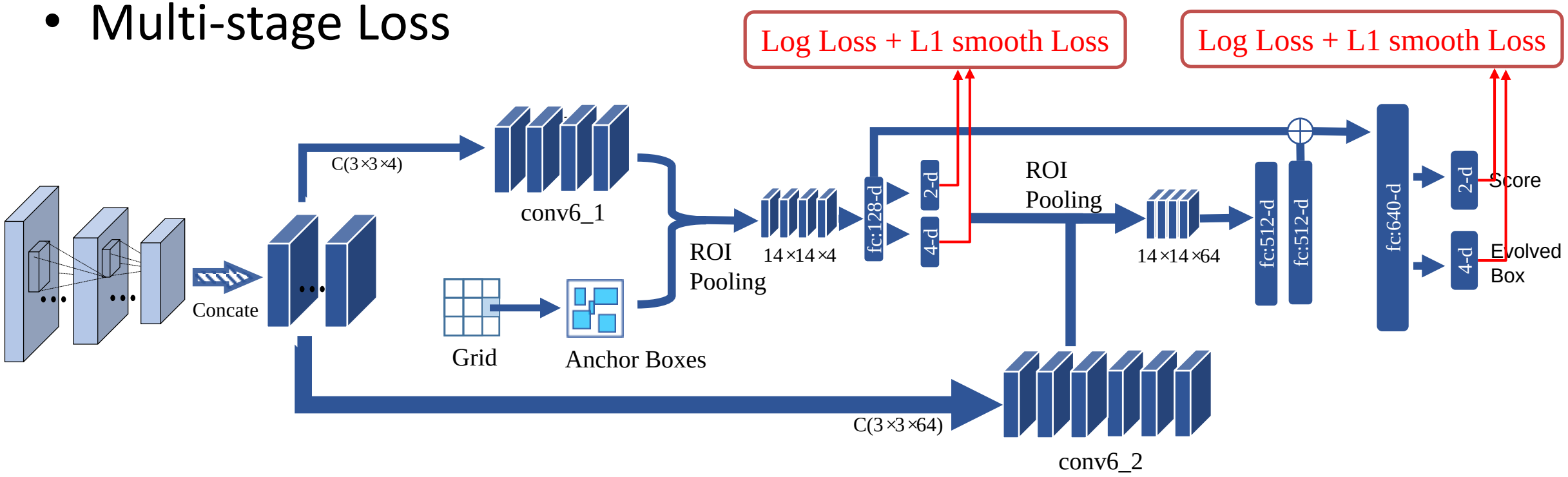

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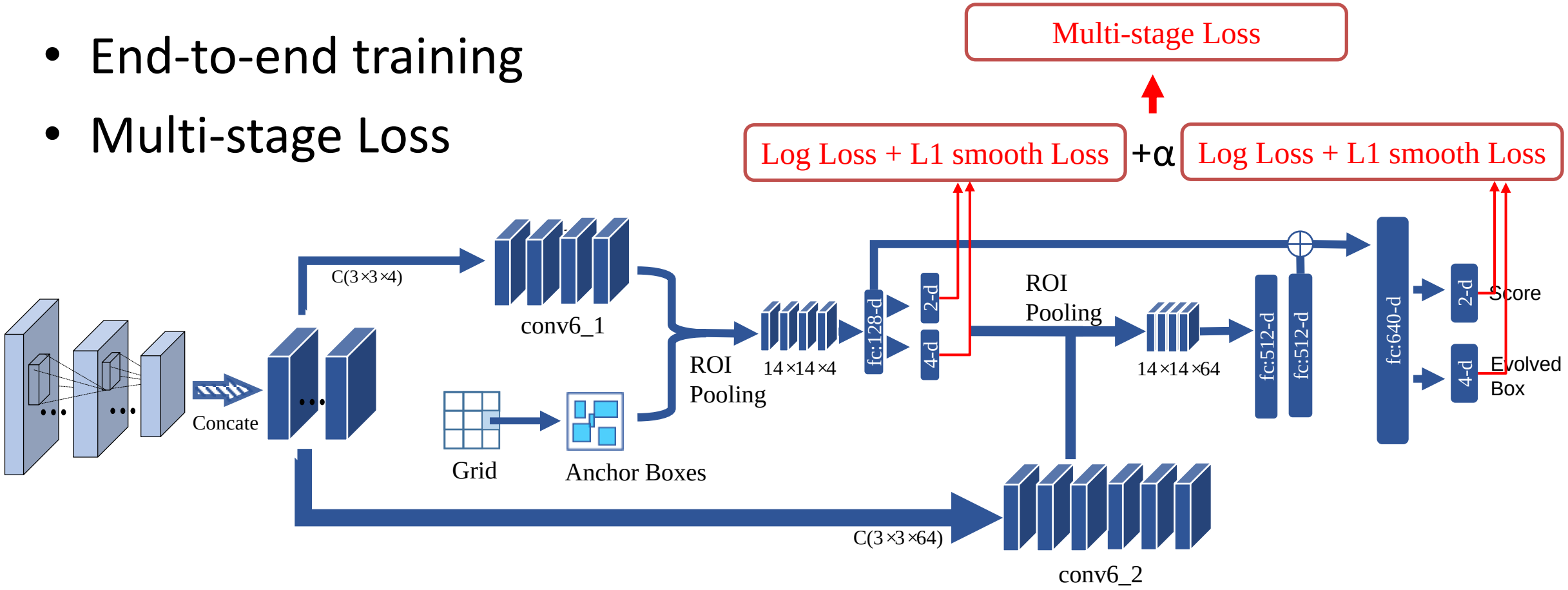

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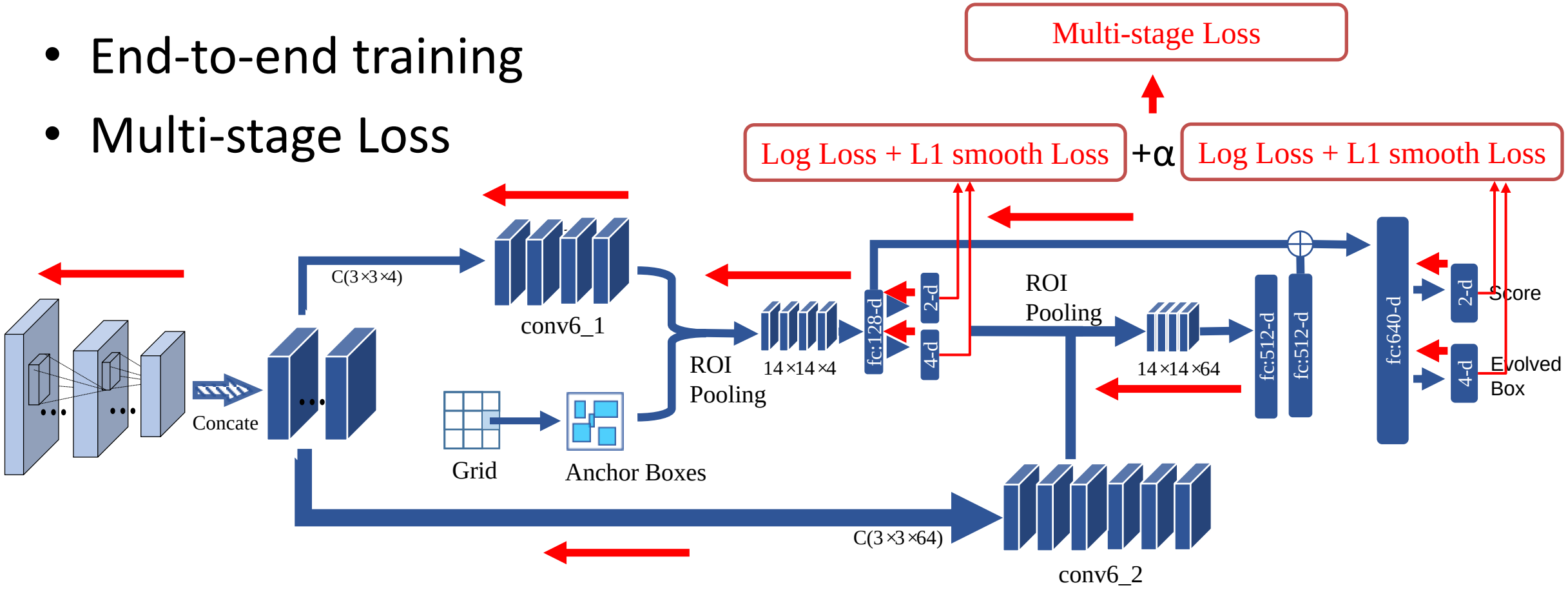

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Experiment

- Dataset
  - DETRAC(Detection)
- Evaluation
  - precision vs. recall (PR)

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Experiment

- Ablation Experiments

Algorithmic Setting	Overall	Sunny	Cloudy	Rainy	Night	Runtime(ms)
PN + FTN + Fusion	73.96	69.70	73.90	82.12	67.91	75
PN + FTN + Concat	83.84	87.09	84.80	85.95	70.21	100
Faster R−CNN[1]	68.58	63.64	70.04	81.56	60.53	87
PN+FTN+Fusion+Concat	**84.43**	**87.48**	**85.88**	**85.65**	**70.86**	110

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Algorithmic Setting	Overall	Sunny	Cloudy	Rainy	Night	Runtime(ms)
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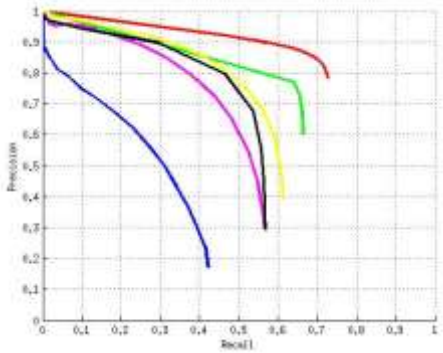
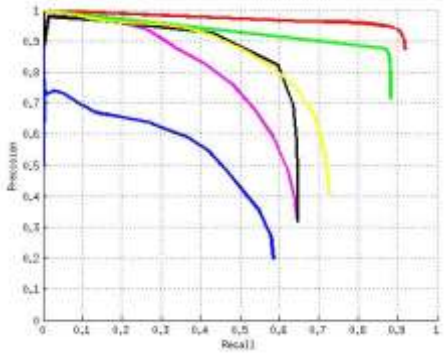
Experiment

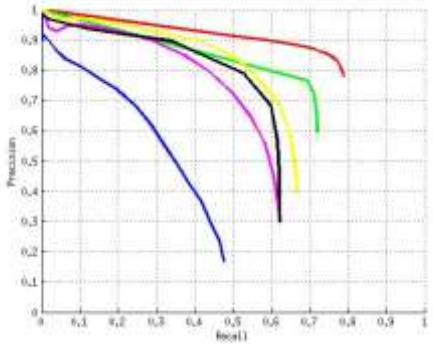
- Comparing with State-of-the-art



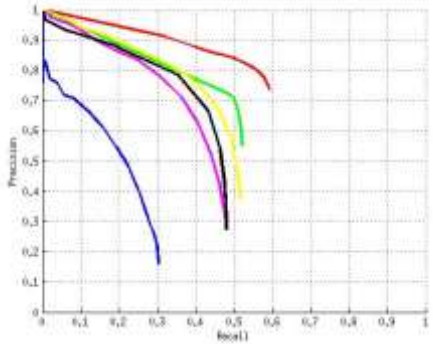
Full



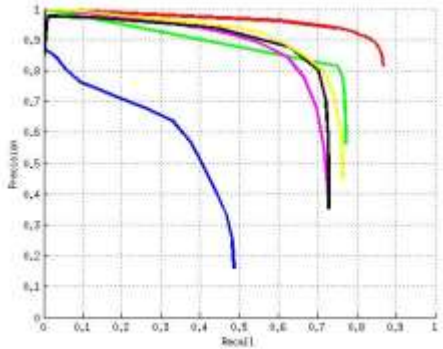
Easy



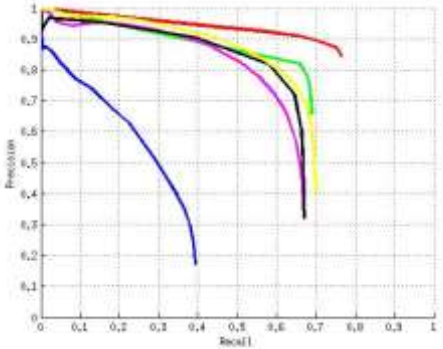
Medium



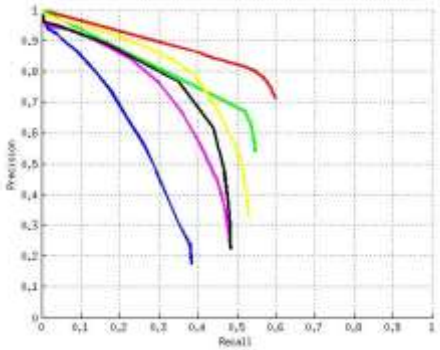
Hard



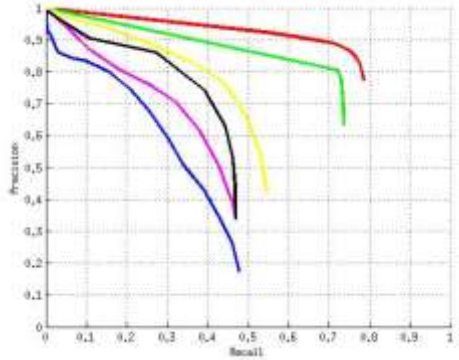
Sunny



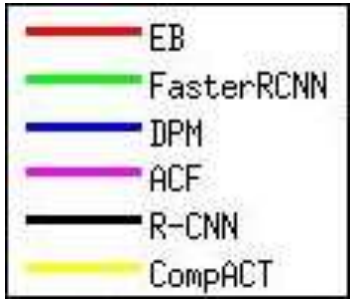
Cloudy



Rainy

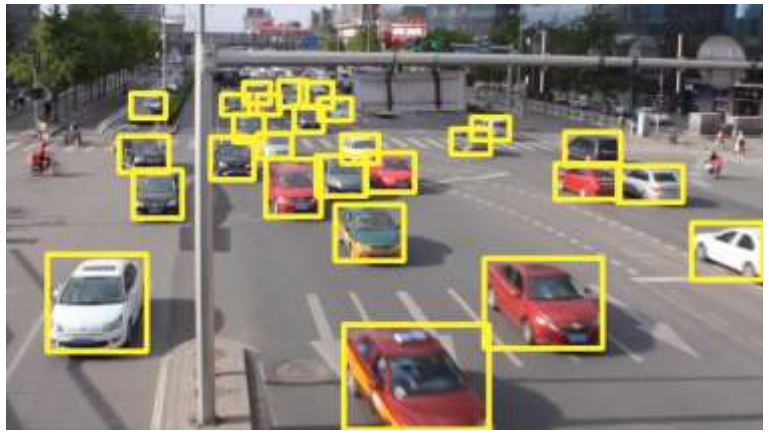


Night

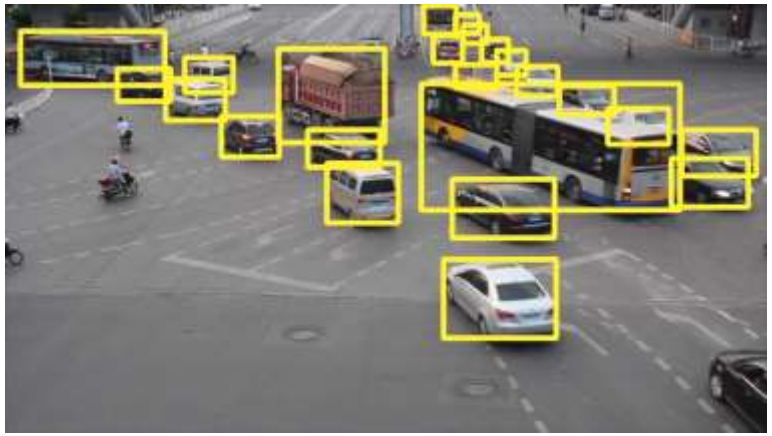


Experiment

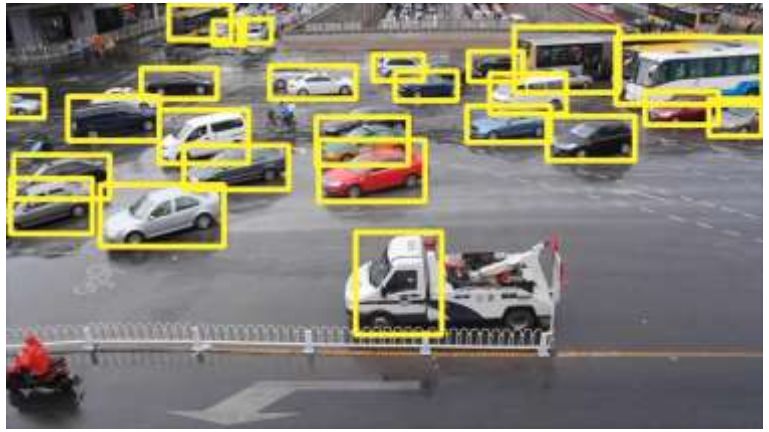
- Results demonstration



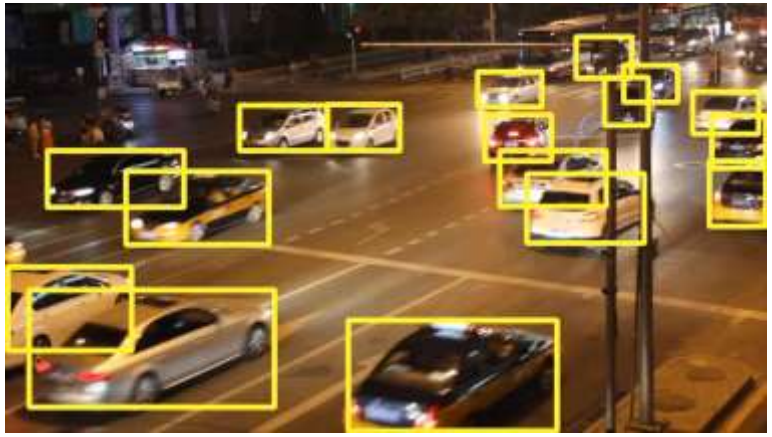
Sunny



Cloudy



Rainy



Night

Conclusion

- The evolving detection framework performs good on DETRAC detection dataset.
- Leverage on different features fusion techniques improves the performance of classification and localization.
- The runtime speed is 9-13 FPS on Titan X (Maxwell).

Thanks!

wangli16@fudan.edu.cn

https://github.com/Willy0919/Evolving_Boxes